

Wazuh Security Monitoring and Alert Detection Report

Submitted By

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Project Title

Implementation of Wazuh SIEM for Linux Log Monitoring and Threat Detection

1. Introduction

This project focuses on the implementation and testing of the Wazuh SIEM platform for monitoring Linux system activity and detecting suspicious behavior.

The objective of this project was to:

- Install and configure a Wazuh server
- Deploy a Wazuh agent on a Linux machine
- Monitor authentication and system logs
- Create custom detection rules
- Detect SSH brute-force attacks
- Analyze alerts generated from suspicious activities

The environment consisted of:

- Wazuh Server
- Linux Agent System
- Kali Linux attacker machine
- SSH service for attack simulation

2. Wazuh Dashboard Monitoring

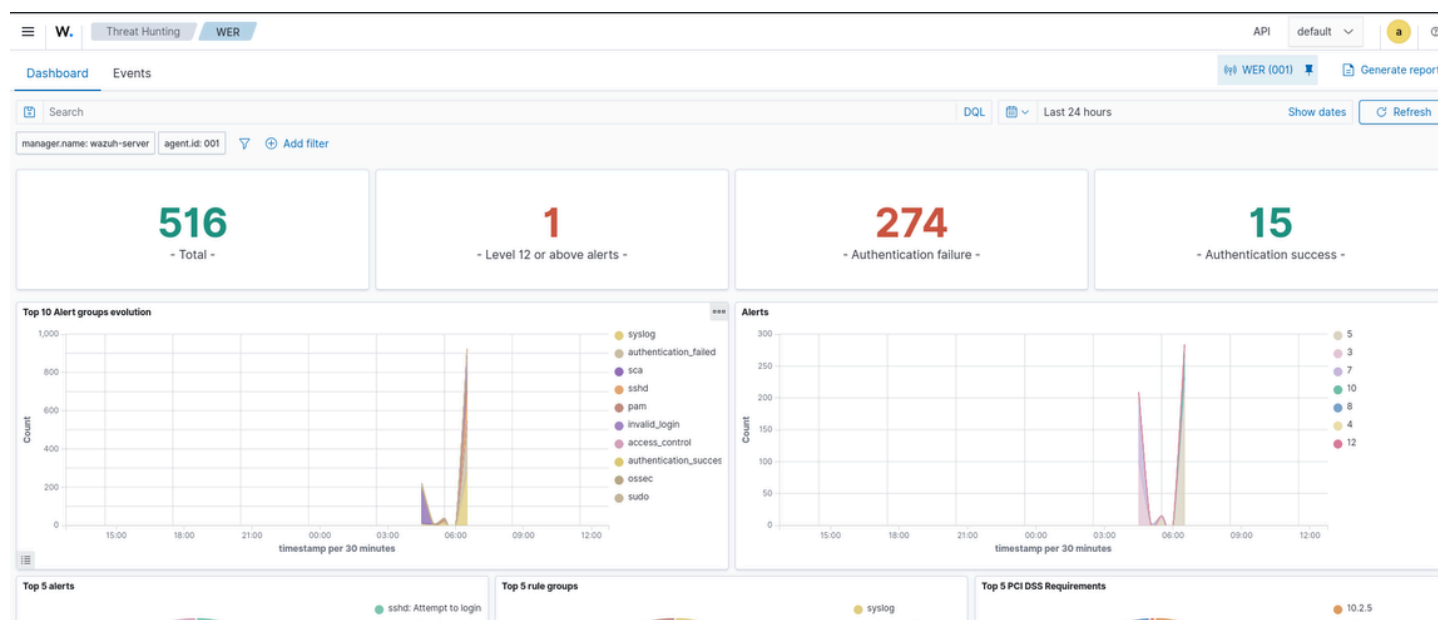
After configuring the Wazuh manager and connecting the agent, the dashboard successfully displayed real-time monitoring data.

The dashboard showed:

- Total security alerts
- Authentication failures
- Authentication success events
- Alert severity levels
- Event timelines
- Top triggered rule groups

From the dashboard:

- Total events collected: 516
- Level 12 or above alerts: 1
- Authentication failures detected: 274
- Authentication success events: 15



The graphs showed spikes in authentication failures during brute-force attack testing.

3. Agent Deployment and Log Collection

A Wazuh agent was installed on the monitored Linux virtual machine.

The agent was configured to collect:

- SSH authentication logs
- sudo command logs
- system authentication events
- journalctl logs

The agent successfully communicated with the Wazuh manager and transmitted logs in real time.

	W.	Threat Hunting	WER	API	default	a
	May 14, 2026 @ 06:34:29.8...	WER	sshd: authentication failed.	5	5760	
	May 14, 2026 @ 06:34:29.8...	WER	sshd: authentication failed.	5	5760	
	May 14, 2026 @ 06:34:27.7...	WER	unix_chkpwd: Password check failed.	5	5557	
	May 14, 2026 @ 06:34:27.7...	WER	unix_chkpwd: Password check failed.	5	5557	
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	May 14, 2026 @ 06:34:27.7...	WER	unix_chkpwd: Password check failed.	5	5557	
	May 14, 2026 @ 06:34:27.6...	WER	unix_chkpwd: Password check failed.	5	5557	
	May 14, 2026 @ 06:34:27.6...	WER	unix_chkpwd: Password check failed.	5	5557	
	May 14, 2026 @ 06:34:27.6...	WER	ssh: authentication failed.	5	5760	
	May 14, 2026 @ 06:34:27.6...	WER	ssh: authentication failed.	5	5760	

4. Custom Rule Creation

To improve threat detection capability, custom XML rules were created inside Wazuh.

These rules were designed to detect:

- **SSH brute-force attacks**
- **Reverse shell commands**
- **Credential harvesting activity**
- **Anti-forensics behavior**
- **Privilege escalation attempts**

5. SSH Brute-Force Detection Rule

A custom XML rule file named ssh-brut.xml was created to detect multiple failed SSH login attempts.

The rule monitors:

- **Failed SSH password attempts**
- **Multiple failures from the same IP address**
- **Repeated authentication failures within a short time period**

Rule Configuration

- **Alert Level: 10**
- **Frequency: 6 attempts**

- **Timeframe: 120 seconds**

ssh-brut.xml

Ruleset Test

Save

```
1<group name="syslog,ssh.brute_force">
2  <!-- Rule for a single failed SSH attempt -->
3  <rule id="100001" level="5">
4    <if_sid>5700</if_sid>
5    <match>sshd: Failed password</match>
6    <description>SSH: Failed password attempt.</description>
7    <group>authentication_failed,</group>
8  </rule>
9
10 <!-- Rule for multiple failed attempts (Brute Force) -->
11 <rule id="100002" level="10" frequency="5" timeframe="120">
12   <if_matched_sid>100001</if_matched_sid>
13   <same_source_ip />
14   <description>SSH: Multiple failed password attempts from same IP.</description>
15   <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
16 </rule>
17 </group>
18
19
```

This rule allowed Wazuh to identify brute-force attacks automatically.

6. Suspicious Command Detection Rule

Another custom XML rule file named command.xml was created to detect suspicious Linux command execution.

The rule monitored commands related to:

- Reverse shells
- Credential harvesting
- Access to /etc/shadow
- Log deletion and anti-forensics commands
- Suspicious shell execution

Examples of Detected Commands

Activity	Example
Reverse shell	bash -i
Shadow access	cat /etc/shadow
Anti-forensics	shred history
Privilege escalation	sudo su

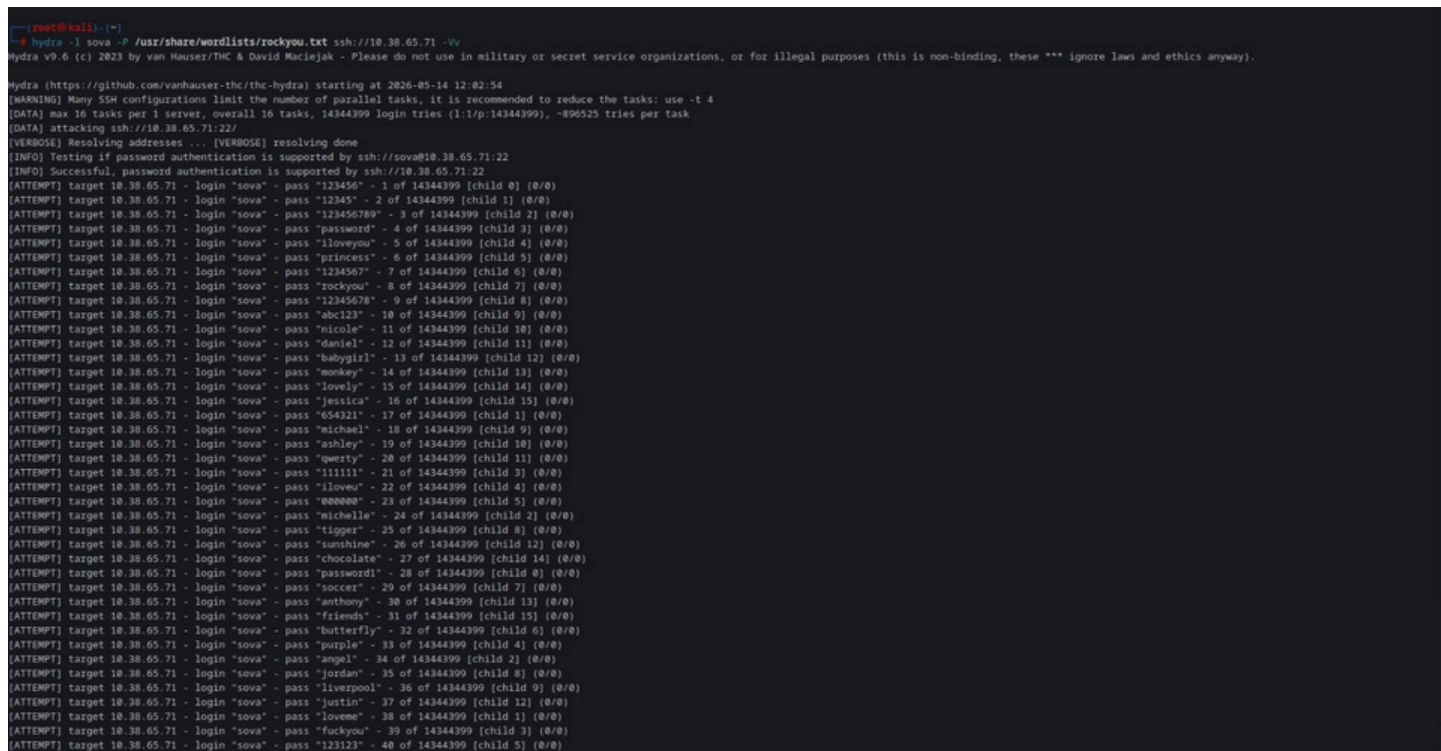


7. SSH Brute-Force Attack Simulation

To test the monitoring system, a brute-force attack was simulated from a Kali Linux machine using the Hydra tool.

Attack

The attack generated multiple failed login attempts against the monitored Linux system.



8. Detection of Authentication Failures

During the brute-force attack, Wazuh successfully detected repeated SSH authentication failures.

The generated alerts included:

- Failed SSH password attempts
- PAM authentication failures
- Multiple failed logins
- Invalid password attempts

The event logs clearly showed repeated attack attempts from the attacker IP address.

	W.	Threat Hunting	WER		API	default		
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	May 14, 2026 @ 06:34:27.6...	WER	unix_chkpwd: Password check failed.		5	5557		
	May 14, 2026 @ 06:34:27.6...	WER	unix_chkpwd: Password check failed.		5	5557		
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	May 14, 2026 @ 06:34:27.6...	WER	sshd: authentication failed.		5	5760		
	May 14, 2026 @ 06:34:27.6...	WER	sshd: authentication failed.		5	5760		
	May 14, 2026 @ 06:34:27.6...	WER	sshd: authentication failed.		5	5760		

Rows per page: 100

< 1 2 3 4 5 6 >

9. Alert Analysis

The generated alerts were analyzed through the Wazuh dashboard.

The alert details included:

- Source IP address
- Username targeted
- Decoder name
- Event timestamp
- Rule description
- Alert severity level

The logs confirmed that Wazuh successfully correlated multiple failed login attempts into a brute-force detection event.

[illegible]

10. Privilege Escalation Monitoring

The system also monitored sudo and privilege escalation activity.

When the command below was executed:

sudo su

Wazuh generated logs showing:

- **Source user**
- **Destination user**
- **Executed command**
- **Terminal session**
- **Timestamp**

The system successfully tracked root access activity.

W. Threat Hunting WER

May 14, 2026 @ 06:47:54.7... WER PAM: Login session closed.

May 14, 2026 @ 06:47:54.7... WER PAM: Login session closed.

May 14, 2026 @ 06:41:32.8... WER PAM: Login session opened.

May 14, 2026 @ 06:41:32.8... WER PAM: Login session opened.

May 14, 2026 @ 06:41:32.8... WER Successful sudo to ROOT executed.

May 14, 2026 @ 06:40:30.2... WER PAM: Login session closed.

May 14, 2026 @ 06:40:21.5... WER Successful sudo to ROOT executed.

May 14, 2026 @ 06:40:21.5... WER PAM: Login session opened.

May 14, 2026 @ 06:40:10.5... WER PAM: Login session closed.

May 14, 2026 @ 06:40:08.2... WER Successful sudo to ROOT executed.

May 14, 2026 @ 06:40:08.2... WER PAM: Login session opened.

May 14, 2026 @ 06:39:20.3... WER PAM: Login session closed.

May 14, 2026 @ 06:39:20.2... WER PAM: Login session opened.

May 14, 2026 @ 06:39:13.7... WER PAM: Login session opened.

May 14, 2026 @ 06:39:13.7... WER PAM: Login session opened.

May 14, 2026 @ 06:39:13.7... WER Successful sudo to ROOT executed.

May 14, 2026 @ 06:38:34.6... WER Multiple authentication failures followed by a success.

May 14, 2026 @ 06:38:34.6... WER PAM: Login session opened.

May 14, 2026 @ 06:34:34.4... WER syslog: User missed the password more than one time

May 14, 2026 @ 06:34:34.3... WER syslog: User missed the password more than one time

May 14, 2026 @ 06:34:34.3... WER syslog: User missed the password more than one time

May 14, 2026 @ 06:34:34.3... WER syslog: User missed the password more than one time

Document Details

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Table JSON

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wazuh-alerts-4.x-2026.05.14

agent.id

001

agent.ip

10.38.65.71

agent.name

WER

data.command

/usr/bin/su

data.dstuser

root

data.pwd

/home/sova

data.srcuser

sova

data.tty

pts/3

decoder.ftscomment

First time user executed the sudo command

decoder.name

sudo

decoder.parent

sudo

full_log

May 14 06:39:34 vbox sudo[131423]: sova : TTY=pts/3 ; PWD=/home/sova ; USER=root ; COMMAND=/usr/bin/su

id

1778721092.236650

input.type

log

location

journalid

manager.name

wazuh-server

predecoder.hostname

vbox

predecoder.program_name

sudo

predecoder.timestamp

May 14 06:39:34

11. Observations

During testing, the following observations were made:

Observation	Result
SSH brute-force attack detection	Successful
Authentication log monitoring	Successful
Custom rule triggering	Successful
Real-time alert generation	Successful
sudo activity monitoring	Successful
Dashboard visualization	Successful

The project demonstrated that Wazuh can effectively monitor Linux systems and detect suspicious activity in real time.

12. Challenges Faced

Challenge	Solution
Agent communication issues	Reconfigured agent settings
XML syntax errors	Corrected rule formatting
Missing authentication logs	Enabled proper log sources
Alert tuning problems	Adjusted rule frequency and levels

13. Conclusion

This project successfully implemented a Wazuh-based SIEM monitoring environment for Linux systems.

The platform successfully:

- Collected logs from a monitored agent
- Detected SSH brute-force attacks
- Generated alerts for suspicious activity
- Monitored sudo and authentication events
- Visualized attack activity through the Wazuh dashboard

The project provided practical experience in:

- SIEM deployment
- Log analysis
- Security monitoring
- Threat detection
- Custom alert rule creation

Wazuh proved to be a powerful open-source platform for real-time security monitoring and incident detection.

14. Future Improvements

Future enhancements may include:

- File Integrity Monitoring (FIM)
- Malware detection
- Active response automation
- Email alert integration
- Windows endpoint monitoring
- Threat intelligence integration

15. References

- [Wazuh Official Website](#)
- [Wazuh Documentation](#)
- [THC Hydra GitHub Repository](#)